EARTHSCAPE CLIMATE AGENCY

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EarthScape Climate Agency

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ACKNOWLEDGEMENT

We express our sincere gratitude to our respected instructor for providing us with the opportunity to undertake this project on **EarthScape Climate Agency**. Their valuable guidance, constructive feedback, and continuous encouragement helped us successfully complete this project.

We also extend our appreciation to our institution for providing the necessary resources and learning environment throughout the course. Special thanks to our teammates for their cooperation, dedication, and teamwork during every phase of the project's development.

Finally, we are grateful to everyone who directly or indirectly contributed to the successful completion of this project.

SYNOPSIS

EarthScape Climate Agency is a web-based Big Data Analytics platform developed to process, analyze, and visualize climate-related information efficiently. The system combines modern web technologies with distributed computing frameworks to manage large volumes of environmental data.

The application enables users to upload climate datasets, process them using **Hadoop**, **MapReduce**, and **Hive**, store information in **MongoDB**, and present analytical insights through an interactive dashboard developed using **React** and **FastAPI**.

The primary objective of the project is to demonstrate how Big Data technologies can be integrated with modern web development to support scalable climate analysis, improve decision-making, and provide meaningful visualizations for users.

ANALYSIS

Climate data is generated continuously from multiple sources such as weather stations, environmental sensors, and research organizations. Due to the increasing volume and complexity of this data, traditional data processing systems often struggle to deliver timely insights.

EarthScape addresses this challenge by implementing a distributed data processing architecture using Hadoop technologies. The system separates data collection, storage, processing, and visualization into independent modules, resulting in improved scalability, maintainability, and performance.

During the analysis phase, the following requirements were identified:

- Secure user authentication and authorization.
- Efficient storage of structured and semi-structured climate datasets.
- Distributed processing of large datasets using Hadoop and MapReduce.
- Data querying using Hive.
- Interactive visualization of processed information.
- Alert generation for significant climate events.
- User-friendly dashboard for monitoring and analysis.

This analysis formed the basis for designing a modular and scalable system architecture.

SCOPE

The scope of the **EarthScape Climate Agency** project is to provide a centralized platform for collecting, processing, analyzing, and visualizing climate-related data using Big Data technologies.

The system covers the following functionalities:

- User authentication and authorization.
- Climate dataset upload and management.
- Storage of climate information in MongoDB.
- Distributed data processing using Hadoop and MapReduce.
- Data querying through Apache Hive.
- Interactive dashboards and graphical visualizations.
- Climate trend analysis and reporting.
- Alert management for environmental conditions.
- User and administrative management.

The current version focuses on providing a functional prototype suitable for academic demonstration. Future versions may include artificial intelligence for predictive climate modeling, integration with real-time weather APIs, GIS-based visualization, cloud deployment, and mobile platform support.

PRODUCT PERSPECTIVE

EarthScape Climate Agency is a standalone web application developed using a modular architecture.

The system consists of a React-based frontend, a FastAPI backend, MongoDB for data storage, and Hadoop technologies for distributed processing.

The architecture separates presentation, business logic, and data layers, making the application scalable, maintainable, and easier to extend.

```
        +----------------------+
        |     React Client     |
        +----------+-----------+
                   |
                   |
           REST API Requests
                   |
                   v
        +------------------------+
        |    FastAPI Backend     |
        +-----------+-------------+
                    |
   +------------------+------------------+
        |                     |
        v                     v
+----------------------+   +----------------------+
|      MongoDB         |   |     Hadoop HDFS      |
|  Application Data    |   |  Distributed Storage |
+----------------------+   +----------+-----------+
                     |
                     v
            +------------------+
            | Hive & MapReduce |
            +------------------+
                     |
                     v
            Analytics & Dashboard
```

OPERATING ENVIRONMENT

The EarthScape application operates within the following environment:

Client Side

- Modern web browser
- Windows 10/11
- Linux
- macOS

Server Side

- Python 3.x
- FastAPI
- MongoDB
- Hadoop
- Hive
- Docker Engine

Development Environment

- Visual Studio Code
- GitHub
- Docker Desktop
- Node.js
- npm
- Python Package Manager (pip)

FUNCTIONAL WORKFLOW

User Login
|
▼
Dashboard
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▼
Upload Climate Dataset
|
▼
Dataset Validation
|
▼
Store in MongoDB
|
▼
Transfer to Hadoop (HDFS)
|
▼
MapReduce Processing
|
▼
Hive Queries
|
▼
Analytics Generation
|
▼
Dashboard Visualization
|
▼
Alerts & Reports

TASKSHEET

Project Title: Earthscape Climate Agency

Project Ref. No: Ep/Advertisement Portal Management System /01		Date Of preparation Of Activity Plan		
S.NO	**TASK**	**ACTUAL START DATE**	**ACTUAL DAYS**	**TEAM MATES NAMES**
1	BACKEND	2 JUNE 2026	30 DAYS	Masoom Mallah
2	FRONTEND			Hamna Sarfaraz
3	LEADING AND MANAGEMENT			Masoom Mallah
4	DOCUMENTATION			Hamna Sarfaraz